1. **Write a C program for Caesar cipher involves replacing each letter of the alphabet with the letter standing k places further down the alphabet, for k in the range 1 through 25.**

Aim:

To write a C program to implement the Caesar cipher by shifting each letter of the alphabet by a key value from 1 to 25.

Procedure:

1. Start the program.
2. Get the plaintext and key value from the user.
3. Shift each alphabet character by the given key.
4. Display the encrypted text.
5. Stop the program

Program (C):

```
#include <stdio.h>
#include <string.h>
int main() {
    char text[100];
    int key, i;
    printf("Enter plaintext: ");
    fgets(text, sizeof(text), stdin);
    printf("Enter key (1-25): ");
    scanf("%d", &key);

    for (i = 0; text[i] != '\0'; i++) {
        if (text[i] >= 'A' && text[i] <= 'Z')
            text[i] = (text[i] - 'A' + key) % 26 + 'A';
        else if (text[i] >= 'a' && text[i] <= 'z')
            text[i] = (text[i] - 'a' + key) % 26 + 'a';
    }

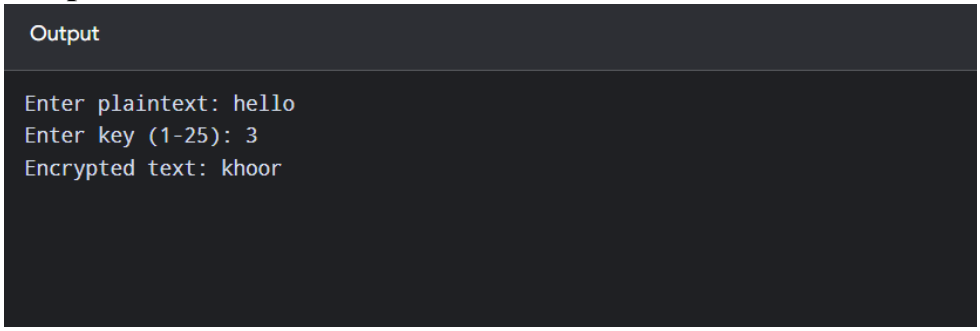
    printf("Encrypted text: %s", text);

    return 0;
}
```

Input:

HELLO
3

Output:



Result:

Thus, the Caesar cipher program was executed successfully, and the plaintext was encrypted using the given key.